Hampel effect (N = 162)
Window coverage (11 frames)
Waveform correlation, r = 0.766
Percentage (%)
100%
80%
60%
40%
20%
0%
17.33%
91.85%
33.05%
67.63%
69.44%
41.7%
99.1%
84.3%
44.8%
64.9%
62.4%
CSI values replaced
Frames with ≥1 replacement
High-change CSI replaced
Temporal variation removed
Peak intensity removed
Static
Walking
Fall
Sleeping
Drinking
Sitting / TV

(a) Aggregate effect and action coverage

Raw
Hampel
Replaced
Normalized amplitude
Time (s)
Static
Walking
3 frames (15 ms)
5 frames (25 ms)
Fall
Sleep
11 frames (55 ms)
21 frames (105 ms)
Drinking
Watching TV
51 frames (255 ms)
101 frames (505 ms)

(b) ElderAL actions (11-frame Hampel)

(c) XRF55 Hampel windows